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AAI EXPERIMENT NO : 07

CODE:

```
import tensorflow as tf
from tensorflow.keras import layers, Model
import numpy as np
import matplotlib.pyplot as plt
# -------------------------------
# 1. Load and Preprocess Dataset
# -------------------------------
(x_train, _), (x_test, _) = tf.keras.datasets.fashion_mnist.load_data()
# Normalize and reshape
x_train = x_train.astype("float32") / 255.
x_test = x_test.astype("float32") / 255.
x_train = np.reshape(x_train, (-1, 28, 28, 1))
x_test = np.reshape(x_test, (-1, 28, 28, 1))
# -------------------------------
# 2. Define Latent Space
# -------------------------------
latent_dim = 2
# -------------------------------
# 3. Encoder
# -------------------------------
encoder_inputs = layers.Input(shape=(28, 28, 1))
x = layers.Conv2D(32, 3, activation="relu", strides=2, padding="same")(encoder_inputs)
x = layers.Conv2D(64, 3, activation="relu", strides=2, padding="same")(x)
x = layers.Flatten()(x)
x = layers.Dense(128, activation="relu")(x)
z_mean = layers.Dense(latent_dim, name="z_mean")(x)
z_log_var = layers.Dense(latent_dim, name="z_log_var")(x)
```

```
# Sampling function
def sampling(args):
    z_mean, z_log_var = args
    epsilon = tf.random.normal(shape=tf.shape(z_mean))
    return z_mean + tf.exp(0.5 * z_log_var) * epsilon
z = layers.Lambda(sampling)([z_mean, z_log_var])
encoder = Model(encoder_inputs, [z_mean, z_log_var, z], name="encoder")
# -------------------------------
# 4. Decoder
# -------------------------------
latent_inputs = layers.Input(shape=(latent_dim,))
x = layers.Dense(7 * 7 * 64, activation="relu")(latent_inputs)
x = layers.Reshape((7, 7, 64))(x)
x = layers.Conv2DTranspose(64, 3, strides=2, padding="same", activation="relu")(x)
x = layers.Conv2DTranspose(32, 3, strides=2, padding="same", activation="relu")(x)
decoder_outputs = layers.Conv2DTranspose(1, 3, activation="sigmoid", padding="same")(x)
decoder = Model(latent_inputs, decoder_outputs, name="decoder")
# -------------------------------
# 5. VAE Model
# -------------------------------
class VAE(Model):
    def __init__(self, encoder, decoder):
        super(VAE, self).__init__()
        self.encoder = encoder
        self.decoder = decoder
    def call(self, inputs):
        z_mean, z_log_var, z = self.encoder(inputs)
        reconstruction = self.decoder(z)
        return reconstruction
    def compile(self, optimizer):
```

```
    super(VAE, self).compile()
    self.optimizer = optimizer
def train_step(self, data):
    if isinstance(data, tuple):
        data = data[0]
    with tf.GradientTape() as tape:
        z_mean, z_log_var, z = self.encoder(data)
        reconstruction = self.decoder(z)
        # Reconstruction loss
        reconstruction_loss = tf.reduce_mean(
            tf.keras.losses.binary_crossentropy(data, reconstruction)
        ) * 28 * 28
        # KL Divergence loss
        kl_loss = -0.5 * tf.reduce_mean(
            1 + z_log_var - tf.square(z_mean) - tf.exp(z_log_var)
        )
        total_loss = reconstruction_loss + kl_loss
    grads = tape.gradient(total_loss, self.trainable_weights)
    self.optimizer.apply_gradients(zip(grads, self.trainable_weights))
    return {
        "loss": total_loss,
        "reconstruction_loss": reconstruction_loss,
        "kl_loss": kl_loss,
    }
def test_step(self, data):
    if isinstance(data, tuple):
        data = data[0]
    z_mean, z_log_var, z = self.encoder(data)
    reconstruction = self.decoder(z)
    # Reconstruction loss
```

```
        reconstruction_loss = tf.reduce_mean(
            tf.keras.losses.binary_crossentropy(data, reconstruction)
        ) * 28 * 28
        # KL Divergence loss
        kl_loss = -0.5 * tf.reduce_mean(
            1 + z_log_var - tf.square(z_mean) - tf.exp(z_log_var)
        )
        total_loss = reconstruction_loss + kl_loss
        return {
            "loss": total_loss,
            "reconstruction_loss": reconstruction_loss,
            "kl_loss": kl_loss,
        }
# -------------------------------
# 6. Train Model
# -------------------------------
vae = VAE(encoder, decoder)
vae.compile(optimizer=tf.keras.optimizers.Adam())
vae.fit(x_train, epochs=10, batch_size=128, validation_data=(x_test, None))
# -------------------------------
# 7. Generate Images
# -------------------------------
def plot_generated_images():
    n = 10
    grid_x = np.linspace(-2, 2, n)
    grid_y = np.linspace(-2, 2, n)
    figure = np.zeros((28 * n, 28 * n))
    for i, yi in enumerate(grid_y):
        for j, xi in enumerate(grid_x):
            z_sample = np.array([[xi, yi]])
```

```
        x_decoded = decoder.predict(z_sample)
        digit = x_decoded[0].reshape(28, 28)
        figure[i * 28:(i + 1) * 28,
            j * 28:(j + 1) * 28] = digit
    plt.figure(figsize=(10, 10))
    plt.imshow(figure, cmap="gray")
    plt.axis("off")
    plt.show()
plot_generated_images()
```

OUTPUT:

```
Epoch 1/10
469/469 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 94s 195ms/step - kl_loss: 3.4725 - loss: 277.8443 - reconstruction_loss: 274.3718 - val_kl_loss: 3.7365 - val_loss: 277.2245 - val_reconstruction_loss: 273.4880
Epoch 2/10
469/469 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 89s 191ms/step - kl_loss: 3.4816 - loss: 262.1476 - reconstruction_loss: 258.6660 - val_kl_loss: 3.7142 - val_loss: 253.8874 - val_reconstruction_loss: 250.1733
Epoch 3/10
469/469 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 84s 179ms/step - kl_loss: 3.4692 - loss: 258.8222 - reconstruction_loss: 255.3530 - val_kl_loss: 3.6170 - val_loss: 252.7298 - val_reconstruction_loss: 249.1129
Epoch 4/10
469/469 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 84s 179ms/step - kl_loss: 3.6485 - loss: 260.5005 - reconstruction_loss: 256.8520 - val_kl_loss: 3.7691 - val_loss: 250.9579 - val_reconstruction_loss: 247.1887
Epoch 5/10
469/469 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 142s 180ms/step - kl_loss: 3.4631 - loss: 265.7708 - reconstruction_loss: 262.3077 - val_kl_loss: 3.5920 - val_loss: 249.1400 - val_reconstruction_loss: 245.5481
Epoch 6/10
469/469 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 84s 180ms/step - kl_loss: 3.5730 - loss: 253.1998 - reconstruction_loss: 249.6267 - val_kl_loss: 3.5716 - val_loss: 248.2174 - val_reconstruction_loss: 244.6458
Epoch 7/10
469/469 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 140s 176ms/step - kl_loss: 3.6830 - loss: 274.3756 - reconstruction_loss: 270.6926 - val_kl_loss: 3.8319 - val_loss: 244.4027 - val_reconstruction_loss: 240.5708
Epoch 8/10
469/469 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 143s 179ms/step - kl_loss: 3.6845 - loss: 253.6175 - reconstruction_loss: 249.9330 - val_kl_loss: 3.6276 - val_loss: 244.0428 - val_reconstruction_loss: 240.4152
Epoch 9/10
```

469/469 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 142s 179ms/step - kl_loss: 3.5466 - loss: 261.4152 - reconstruction_loss: 257.8686 - val_kl_loss: 3.7487 - val_loss: 243.7393 - val_reconstruction_loss: 239.9906

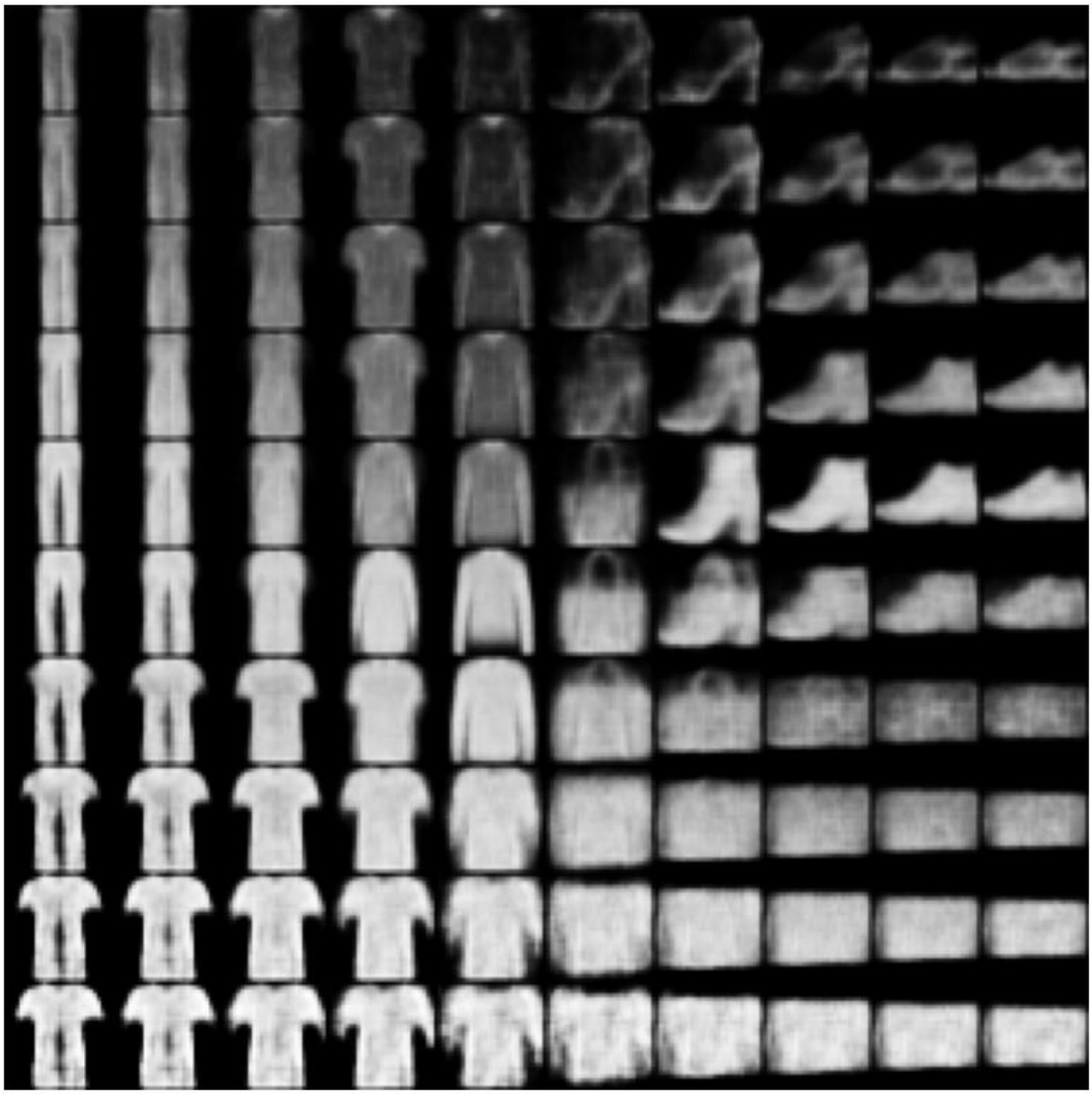
Epoch 10/10

469/469 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 84s 180ms/step - kl_loss: 3.6997 - loss: 260.7805 - reconstruction_loss: 257.0808 - val_kl_loss: 3.8567 - val_loss: 245.7294 - val_reconstruction_loss: 241.87271/1 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 61ms/step

1/1 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 60ms/step

1/1 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 70ms/step

1/1 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 61ms/step

1/1 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 65ms/step

1/1 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step